Bases & the Steinitz $\boxed{\text{exchange}}$ lemma

Lemma:

V : F - vector space

$Y, X \subseteq V$: finite subsets

s.t. (a) $\underbrace{\text{Span}(X) = V}$

(b) Y is $\underbrace{\text{linearly independent}}$

Suppose that Y has m elements

& X has n elements

Then : ① $m \leq n$

② If $Y = \{y_1, y_2, \dots, y_m\}$

then we have $X = \{x_1, x_2, \dots, x_m, x_{m+1}, \dots, x_n\}$

s.t. the set $X' = \{y_1, y_2, \ldots, y_m, x_{m+1}, \ldots, x_n\}$

also spans V (i.e. $\mathrm{Span}(X') = V$)

Corollary ①: Suppose that V is an F-vector space & $\exists$ a finite subset $X \subseteq V$ s.t. $\mathrm{Span}(X) = V$. Then:

ⓐ V admits a basis

ⓑ Every basis for V has the same number of elements

Pf:

(a) We need to show:

There is a subset $Y \subseteq V$

s.t. • $\operatorname{span}(Y) = V$

• Y is linearly independent.

$P(n)$: If V is an F-vector space spanned by a subset with n elements, then V admits a basis with finitely many elements

Base case ($n=1$)

$$V = \operatorname{Span}(\{v\}) = \{c \cdot v : c \in F\}$$

• If $v \neq 0$ then $\{v\}$ is linearly independent & so is a basis

- If $v = 0$, then $V = \{0\}$, then $\emptyset$ is a basis.

Inductive hypothesis:

$P(n-1)$ is true.

Inductive step:

V is spanned by $X = \{v_1, \ldots, v_n\}$

- If X is linearly independent then X is a basis & we're done
- If X is $\boxed{\text{not}}$ linearly independent then (after reindexing the subscripts) we find that v_n is a linear

combination of $\{v_1, v_2, \ldots, v_{n-1}\}$

In this case,

$$\mathrm{Span}(\{v_1, v_2, \ldots, v_{n-1}\}) = \mathrm{Span}(\{v_1, v_2, \ldots, v_n\})$$
$$= V$$

$\Rightarrow$ V can be spanned by $n-1$ elements

$\xRightarrow[\text{is true}]{P(n-1)}$ V admits a basis with finitely many elements

(b) Suppose that X_1, X_2 are two bases for V and suppose that X_1 is finite.

WTS:
- X_2 is finite
- X_2 has the same number

of elements as X_1

We will use the Exchange lemma

If $\underbrace{\{x_1, x_2, \dots, x_n\}}^{\text{linearly ind.}} \subseteq X_2$

and $\underbrace{\{y_1, \dots, y_m\}}_{\text{Spans } V} = X_1$

$\xrightarrow[\text{lemma}]{\text{Exchange}}$ $n \leq m$

So every finite subset of X_2 has no more than m elements

$\Longrightarrow$ X_2 is finite with no more than m elements

$\Rightarrow$ X_2 is a finite basis with no more elements than X_1

But now we can flip the roles of X_1 & X_2 and we find that

X_1 has no more elements than X_2

$\Rightarrow$ X_1 & X_2 have the same number of elements.